

Mandeep

✉ mandeepjangir@proton.me *Junior Research Fellow at (Climate & research Services)*  mandeepairborne.github.io
India Meteorological Department
Pune, India

Interested in atmospheric physics, weather dynamics, and climate science. Focus on using numerical models, observational data, and machine learning to study wind patterns, wave propagation, and monsoon processes.



Skills

- | | | |
|--------------------------|--------------------------|-------------------------|
| * Languages | * Libraries | * Software |
| > Python | > ROOT | > Linux (Ubuntu & Arch) |
| > C++ | > SciPy | > Git |
| > Bash | > NumPy | > Jupyter Notebooks |
| > LaTeX | > Pandas | > Overleaf |
| | > Matplotlib | |
| | > TensorFlow | |
| | > Keras | |
| | > Mathematica | |

Experience

- * **Junior Research Fellow** • 2025 – Present
India Meteorological Department (IMD) • Climate & research Services • Pune, India
 - > Researching atmospheric waves, mountain-forced winds, and weather patterns over complex terrains.
 - > Using machine learning and Python to analyze weather and climate datasets.

Education

- * **MSc in Physics** • Aug '22 – May '24
Dr. B R Ambedkar National Institute of Technology • Jalandhar, India
- * **BSc in Physics, Chemistry, Mathematics** • June '19 – July '22
St. Aloysius College (Autonomous) • Mangalore, India

Research Projects¹

- * **Numerical Modeling of Advective Boundary Layer Dynamics** • May '26
ICTS Monsoon School Project • Bengaluru, India
 - > Built a numerical solver for the Mahrt (1972) boundary layer model using Python.
 - > Initialized the model with ERA5 wind data along the 60°E longitude (Somali Jet axis).
 - > Simulated wind profiles and studied boundary layer sensitivity to pressure forces and eddy viscosity.
- * **Autoencoder-based Precision Measurement Techniques** • July – Nov '25
Northwestern University (USA) • Supervisor: Dr. Raman Khurana
 - > Built autoencoder neural networks to improve precision in particle physics measurements.
 - > Used the models to calibrate measurements and reduce systematic uncertainties.
- * **Medium Modification of Pentaquark States** • July '23 – May '24
Dr. B R Ambedkar NIT Jalandhar • Master's Thesis Research
 - > Studied exotic pentaquark particles in dense stellar matter using quark field models.
 - > Calculated mass modifications in nuclear and strange matter relevant to neutron star cores.
 - > Used scalar and vector meson frameworks for dense matter simulations. [Link to thesis excerpts.](#)
- * **Galaxy Morphology Classification using Deep Learning** • May – July '23
Independent Research Project
 - > Built a convolutional neural network (CNN) to classify galaxy shapes using SDSS image data.
 - > Created a data preprocessing and augmentation pipeline for astronomical images.
 - > Evaluated classifier performance across multiple morphological classes. [Project details.](#)

Training & Summer Schools

- * **Participant** • May '26
International Centre for Theoretical Sciences (ICTS) • Moist Convective Dynamics of Monsoons - II • Bengaluru, India

¹Scan the QR code or visit the website for project codes and details: <https://mandeepairborne.github.io/>

* **Participant** • Winter '25

Brown University Winter AI School • AI trends & applications in scientific computing • Rhode Island, USA

* **Participant** • Summer '22

Curves and Surfaces • Differential geometry and applications in theoretical physics • ICTS, Bengaluru, India

* **Participant** • Summer '21

Physics Training and Talent Search (PTTS) • Advanced undergraduate physics enrichment program • IISER TVM, India

Selected Courses

- * **Master's Courses:** Quantum Mechanics, Relativistic Quantum Mechanics, Particle Physics, Nuclear Physics, Electrodynamics, Statistical Mechanics, Mathematical Methods.
- * **BSc Courses:** Classical Mechanics, Modern Physics, Thermodynamics, Distribution Theory, Linear Programming & Optimization.
- * **Self-Paced / Additional Courses:** Machine Learning with Python, Deep Learning Specialization.

Awards & Achievements

- * **CSIR-NET (Physical Sciences) – AIR 143:** Qualified for Assistant Professor appointment and PhD admission (Dec 2024).
- * **IIT JAM – AIR 2283:** Qualified joint entrance exam for Master's programs (India).

Leadership & Student Governance

* **Postgraduate Student Council Representative** • 2023 – 2024

Department of Physics, NIT Jalandhar • India

- Represented postgraduate physics students in the Institute Senate on academic curriculum and policy reform.

* **C-Certificate Cadet** • 2019 – 2022

National Cadet Corps (NCC) – Air Wing • St. Aloysius College, Mangalore

- Participated in leadership camps and studied aviation physics and basic aerodynamics.

References

Dr. Soumi Chakravorty (PhD Supervisor)

Scientist-D

India Meteorological Department, Pune

Email: soumichakravorty.imd@gmail.com

Scholar Profile: Google Scholar

Dr. Suneel Dutt (Master's Thesis Supervisor)

Assistant Professor

Department of Physics, NIT Jalandhar, India

Email: dutts@nitj.ac.in

Scholar Profiles: NIT Jalandhar | Google Scholar